



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No: MEL G627
Course Title: VLSI Fabrication Technology
Instructor-in-Charge: Dr. Satyendra Kumar Mourya
Lab Instructors: Mr. Nilesh Jaiswal (RS), Mr. Pradeep Verma (RS)

1. Course Description:

Historical perspective; Processing overview; Crystallography of semiconductors; Defects in semiconductors; Clean room; Wafer- growth; cleaning; oxidation; epitaxy; wet and dry etching; lithographic process; diffusion; ion implantation; deposition of films using CVD, and PVD techniques; device and circuit fabrication; Process Integration: Passive components, Bipolar Technology and MOSFET Technology, Fabrication of advanced VLSI devices; System on chip; Future trends and Challenges:

Laboratory components: Simulation of key fabrication processes using CAD tools; Demonstration of basic FEOL/BEOL fabrication processes; Fabrication and testing of electronic devices.

2. Scope and Objective of the Course:

The objective of the course is not only to introduce the basic principles involved in IC fabrication, but also, to discuss processing technology. Fabrication of integrated circuits is a joint venture by electrical engineers, chemical engineers, materials scientists, and physicists. This interdisciplinary course builds bridges across various disciplines.

3. Text Books: S. M. Sze, VLSI Technology, TATA McGraw-Hill.

4. Reference Books: 1) Plummer, Deal, and Griffin, Silicon VLSI Technology, Pearson.

2) May and Sze, Fundamentals of Semiconductor Fabrication, Wiley.

3) William D. Callister and Jr. David G. Rethwisch, Materials Science and Engineering: An Introduction, Wiley.

4) S.A. Campbell, The Science and Engineering of Microelectronic Fabrication, Oxford.

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1. Introduction and Overview	3	Introduction-(T), Chapter-1 (R3), Lecture slides/Notes	A brief overview, identifying the unit IC fabrication steps by studying BJT and MOSFET fabrication processes
2. Crystal growth, Crystal defects, Clean room, and wafer preparation	6	Chapter-1- (T), Chapter-3 (R3), Lecture slides//Notes	Crystal structure of Si, Concepts of Crystal structure, Miller indices, orientation, defects, Introduction to clean room, Clean room basics, Clean room attire, Standards of cleanrooms,



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			Crystal growth techniques: Czochralski and FZ growth methods, Wafer preparation and specifications
3 Unit Fabrication Steps			
3.1 Oxidation	5	Chapter-3-(T), Chapter-6-(R3), Lecture slides/Notes	Knowledge of Oxidation techniques and systems, growth kinetics, various applications of SiO ₂ in BJT, and MOS processes.
3.2 Epitaxy	3	Chapter-2-(T), Chapter-14-(R1), Lecture slides/Notes	Epitaxial growth processes, equipment and doping, and autodoping.
3.3 Diffusion	4	Chapter-7 -(T), Chapter-3-(R1), Lecture slides/Notes	Diffusion models and types, equipment, and applications.
3.4 Ion Implantation	4	Chapter-8-(T), Chapter-5-(R1), Lecture slides/Notes	Ion implantation models, equipment, and applications.
3.5 Lithography	4	Chapter-4 (T), Chapter-5-(R3), Lecture slides/Notes	Photolithography, Electron beam, X-ray, and Ion lithography.
3.6 Etching	4	Chapter-5 (T), Chapter-10 - (R3), Lecture slides/Notes	Physical and chemical etching, RIE.
3.7 Deposition/Metallization	4	Chapter-6 (T), Chapter-9 - (R3), Lecture slides/Notes	Deposition (physical) - DC, RF, and magnetron sputtering and their application. Deposition (chemical) - CVD, APCVD, LPCVD, and PECVD and their application.
4. Fabrication of advanced VLSI devices; System on chip; Future trends and Challenges	3	Lecture slides/Notes and research papers	Fabrication of GAA, FinFET, TFET, and other devices, as well as current trends in semiconductor industries.
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6. List of Lab Experiments (Tentative)

S.No.	Experiments	Location
1.	Familiarization with the Fabrication lab, its protocol & clean room classifications.	SMSD Lab
2.	Wafers and Dicing: Phase identification, XRD, lattice planes, importance of phases, and cleaning protocol.	SMSD Lab
3.	Deposition Techniques and Characterization: SMO thin film deposition + Annealing.	SMSD Lab
4.	Real time gas sensing measurements and IV analysis of SMO Thin Film.	SMSD Lab
5.	To study thin film patterning using lithography technique.	SATHI Foundation
6.	To study thin film growth using the sputtering technique.	SATHI Foundation
7.	Introduction to TCAD and Process Simulation.	O-Lab

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Closed Book/ Open Book)
Quizzes	-	15%	TBA	Open book
Seminar	-	5%	TBA	Open book
Assignments	-	10%	TBA	Open book
Mid-semester Test	90 minutes	20%	TBA	Closed Book
Comprehensive Examination	180 minutes	40%	TBA	Open Book + Closed Book
Lab (Viva + Quiz)	-	10%	TBA	Closed book

8. Chamber Consultation Hour: To be announced in the class.

9. Notices: Notices, if any, will be displayed on the EEE Notice Board// NALANDA (online portal).

10. Make-up Policy: Make-up will be given only to genuine cases. Applications for make-up should be given in advance, and prior permission should be obtained for scheduled tests

Instructor-in-charge
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